

Lecture 3

Specifications, SDD, Testing, TDD, Lab 2

Suthep “Jogie” Madarasmi, Ph.D.

Weeks Taught by Aj. Jogie

Date	Topics	Activities
W3. 18 Aug	Specification, Architecture, Tech Stack, Data Modeling, and Lightweight UML	Lab 2-1. Ticket Creation: engineering contract, Spec DD, Test DD, DB increment & seed, REST API contract
W4. 25 Aug	Test Design, TDD, Verification, Automated Testing, and Definition of Done	Lab 2-2. Create Ticket / My Tickets / Ticket Detail screens, attachments, Zen Green theme UI, full test suite (unit/API/UI/E2E)
W10. 13 Oct	Midterm Review and Individual Practical Integration. (Holiday)	Lab: Individual Practical Challenge & Demo (midterm integration); begin Team Sprint 1: working vertical slice
14–28 Oct	Midterm Exam (covers Weeks 1–7)	
W11. 3 Nov	Cloud Deployment and Delivery to Users	Lab: Cloud Deployment Lab. Deploy the team application to the cloud (GCP) and deliver to users; Team Sprint 2
W13. 17 Nov	Metrics, Reviews, Retrospectives, Ethics, Responsible AI Use, and AI Usage Report	Lab: Metrics, code review & retrospective; Responsible-AI reflection and AI Usage Report; final project hardening
1–11 Dec	Final Exam (covers Weeks 8–14)	After Finals: Project Demo.

Expectations of a Capable Software Engineer

Even if no LLM existed, a good software engineer should be able to produce:

1 Requirements

2 Specifications

3 Design Contracts

4 Acceptance Criteria

5 Test Plan

6 Implementation (Coding) Plan

The Role of AI

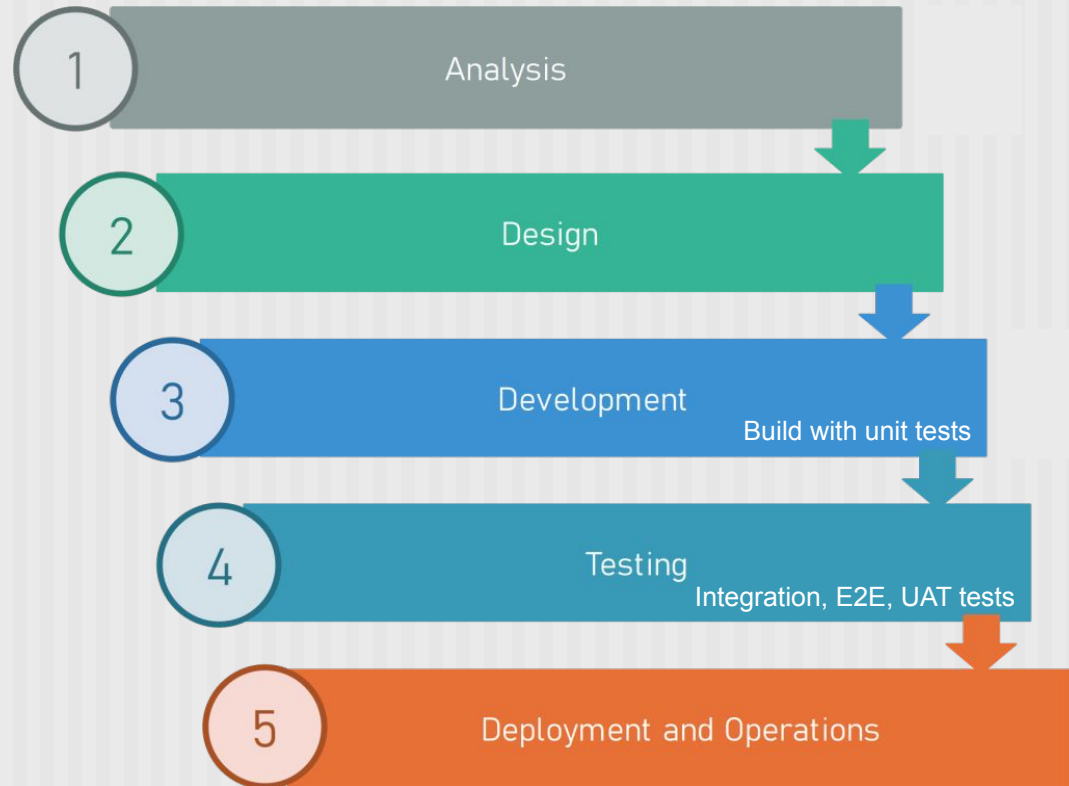
The LLM is simply a tool that can help engineers produce and refine them faster.

Software Development Life Cycle (SDLC):

SDLC #1. Waterfall Model

Classic, but the reality is

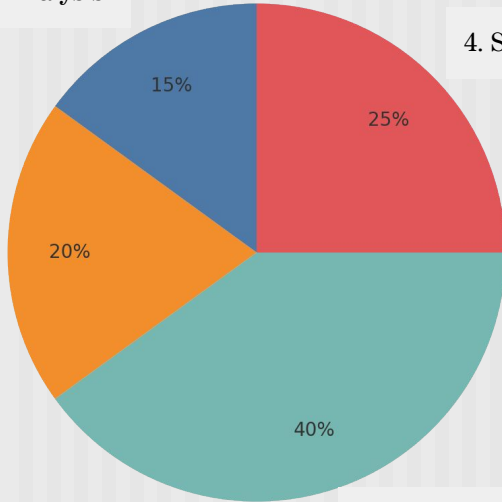
1. Activities are not sequential
2. Activities can be worked on in parallel
3. It is necessary to loop back to an "earlier" activity



Rough Effort (mandays) vs. Timeline (calendar days)

Man-days Effort Distribution

1. Requirements / Analysis



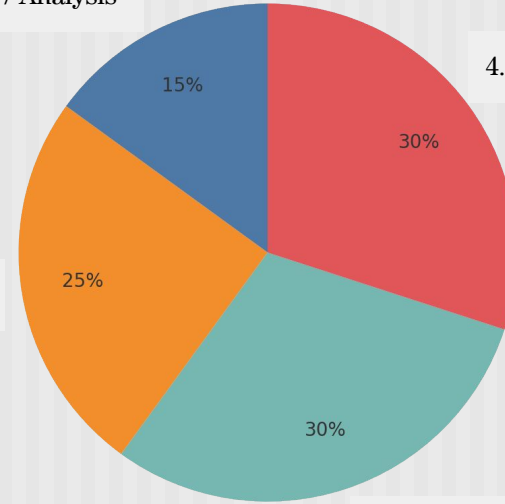
4. SIT, E2E, UAT Tests

2. Design

3. Coding & Unit Test

Calendar-days Effort Distribution

1. Requirements / Analysis

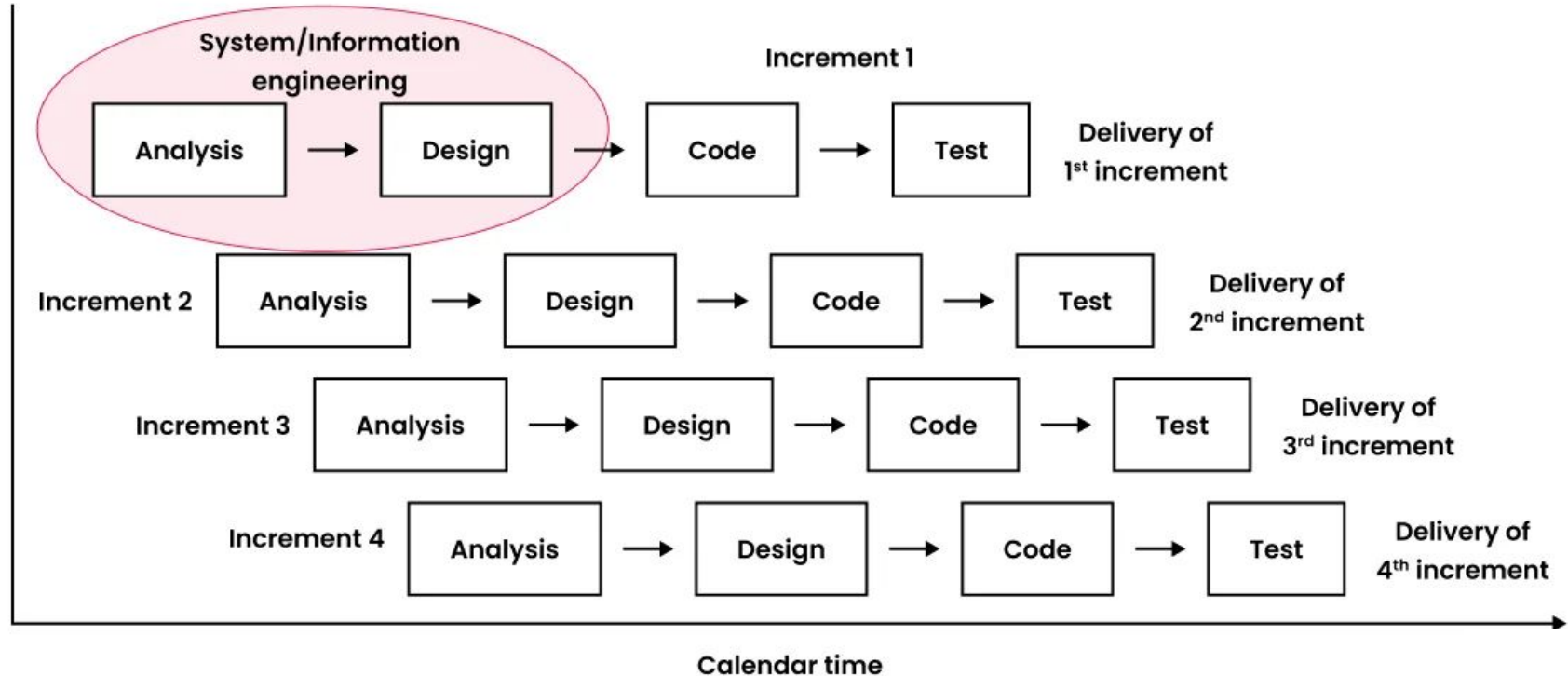


4. SIT, E2E, UAT Tests

2. Design

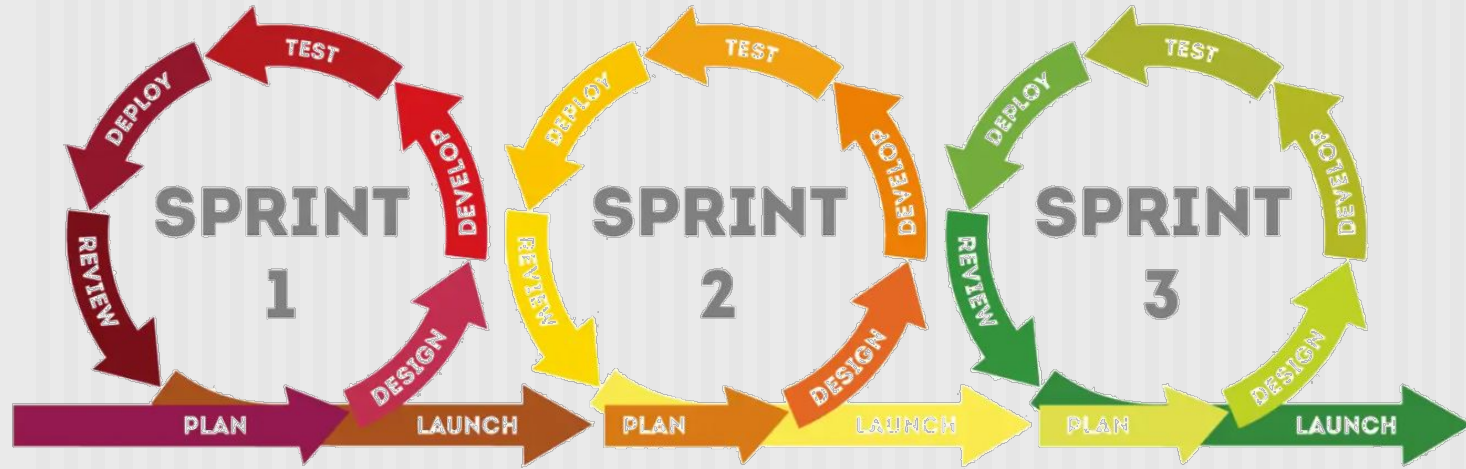
3. Coding & Unit Test

SDLC #2: The Incremental Model. **Most Common.**



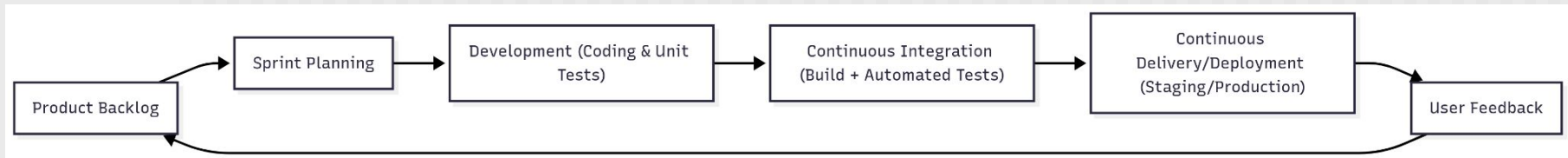
SDLC #3: The Agile Model.

Focus on “Working software over detailed documentation”

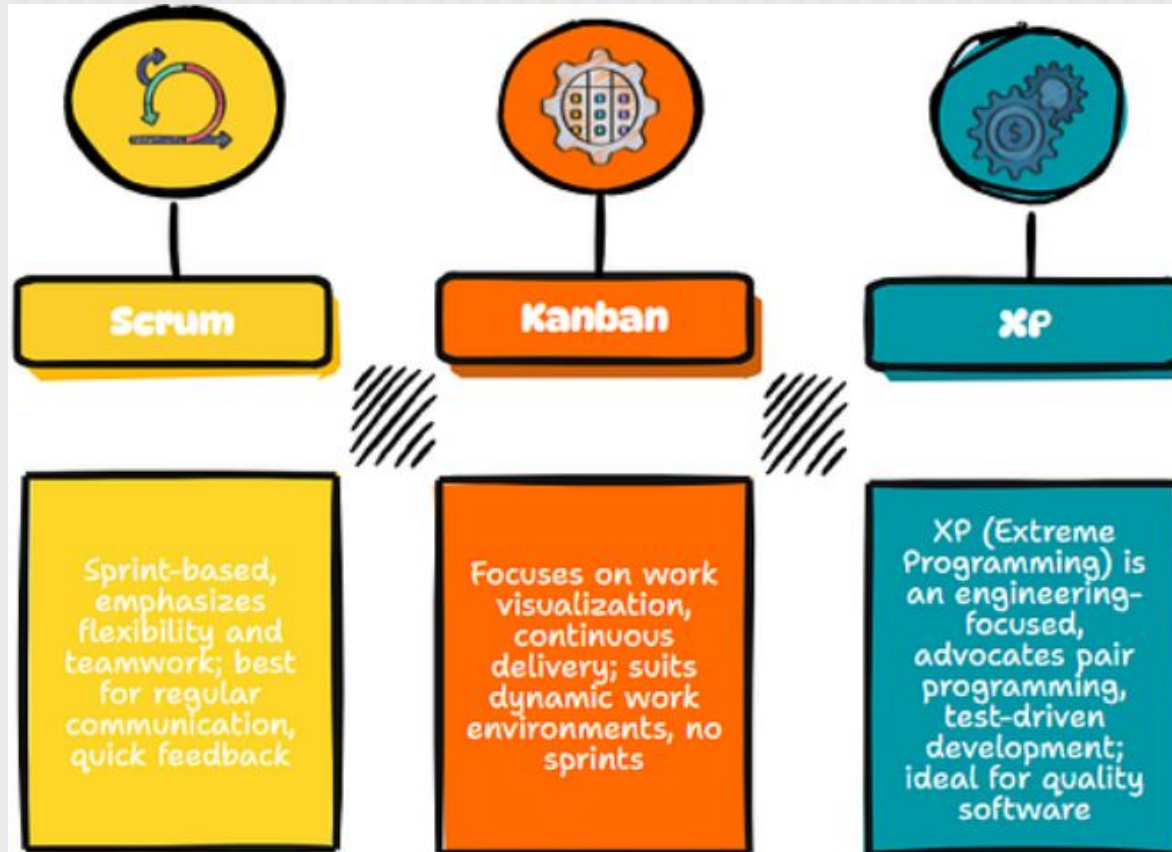


Agile values rapid iteration → **CI/CD** enables frequent integration, testing, and delivery without bottlenecks.

- **Continuous Integration (CI)** ensures code from multiple developers is merged often with automated tests.
- **Continuous Delivery/Deployment (CD)** automates packaging and release, with quick deployment to production.



Top 3 Agile Frameworks



Lab 2. TokTickIT

Sprint 2

Lab 1. Full Stack Hello World Starter.



CPE 334 Introduction to Software Engineering in the Age of AI Agents

Sections 1, 2, HS, 31 and 32. Semester: 1/2026.

Lab 1. TokTickIT (ตอกตึกกิต) Full-Stack Hello World Starter.

Score: ____ / 40

3. Lab 1 Request from Stakeholder

Here the stakeholder refers to the product owner, customer, or end user requesting this application.

“Before developing features for this ticketing app, the IT department wants evidence that the proposed technology stack works as one integrated system. Create a basic TokTickIT application that displays the service status and the supported request categories stored in the database.”

TokTickIT IT Service Desk

[Check System]

System Status: Online

Supported Request Categories:

- Account and Access
- Hardware
- Software
- Network

Lab 2. Requester Ticketing MVP with UI Foundations

CPE 334 Introduction to Software Engineering in the Age of
Sections 1, 2, HS, 31 and 32. Semester: 1/2026.

Lab 2. TokTickIT Requester Ticketing MVP with UI Foundations

Figure 1. Illustrative TokTickIT Ticket Detail screen using the Zen Green template design language. Lab 2 implements the Requester-facing Create Ticket, My Tickets, Ticket Detail, and Attachment functions. Later labs will add role-specific IT Staff controls, communication, workflow, and Actions Taken.

TokTickIT My Tickets Create Ticket Profile

My Tickets > Ticket Details Back to My Tickets

Ticket No. TKT-2025-001234	Ticket Date May 12, 2025 09:14 AM	Category Hardware	Related System Corporate Laptop
Requester Jennifer Anderson	Requested Priority Medium	IT Priority Medium	Current Status In Progress
Ticket Owner Michael Brown (IT Support)	Summary Laptop battery drains quickly		

Description

My laptop battery is draining much faster than usual even when the system is idle. This started happening after last week's Windows update.

Resolution Summary

No resolution summary available yet.

Public Comments 3 Attachments 2 Service Actions 1 Event Log 6

Add Comment

Type your comment here... Post Comment

JA Jennifer Anderson Requester May 13, 2025 11:45 AM
Thank you for the update. Please let me know if you need any additional information.

MB Michael Brown IT Support May 13, 2025 10:30 AM
We are investigating the issue on your device. We'll update you shortly.

JA Jennifer Anderson Requester May 12, 2025 09:20 AM
Just adding that this issue occurs even when I close all applications.

Main Example used in my Lectures: POS System

Requirements List (minimalist)

1. **Serve customers at the counter**
 - Cashiers can start a checkout, add or remove items, apply valid discounts, and see the running total.
 - A sale can be **cancelled** any time before it is finished.
2. **Accept payments**
 - Customers can pay **by cash** or **by credit card** which is processed via a **Payment Service Provider** (eg., K-Bank)
 - The system **works with the card machine** from, say, K-Bank to send the amount and receive approval
 - If a card is **declined**, the cashier can retry, switch to cash, or cancel the sale.
3. **Print documents: Receipt** for successful sale, **Refund Receipt** for refunds. Prints to store **receipt printer**.
4. Updates stock in **store's inventory service** app: **reduces** stock of the sold item, adds back stock of returned items.
5. **Handle returns and refunds.**
 - Cashiers or managers can find the original sale and **process a return**.
 - The system **refunds** the customer (cash or card, following store policy) and records the reason.
6. **Manager Functions**
 - Managers can view a **daily sales summary** for a selected date, including total cash & card amount received.
 - Managers can **add or update products and prices** to be used by POS.

Example Core Scenarios

1. Place Order / Checkout
2. Pay by Cash
3. Pay by Card (with Authorization)
4. Cancel a Sale (Midway)
5. Process Return / Refund
6. Print Receipt
7. Update Stock for Sales & Returns

Possible Additional Functionality to POS

- **Point-of-sale operations:** price override/manager approval, split tenders/partial payments, tips/gratuities, gift cards/vouchers, coupons/promotions, receipt reprint, cash drawer open/close, no-sale.
- **Customer & loyalty:** customer lookup, loyalty points, membership pricing.
- **Inventory/admin:** stock receive/transfer/adjust, product & price management, tax/VAT rules, end-of-day reconciliation (Z-report), shift management.
- **Resilience:** offline mode (store-and-forward), retry/capture later.
- **Security/audit:** login/roles, audit trail.
- **Multi-store:** branch configuration, central reporting.

POS UI Design for POS Order Form View

POS Terminal

Cashier: Jane

Shift: 09:00–17:00

Add Item by Scan/Search here...

Item	Price	Qty	Line Total
Pencil SKU: STN-001	฿20.00	- 10 +	฿200.00 ×
A4 Paper SKU: STN-002	฿100.00	- 2 +	฿200.00 ×
Subtotal			฿400.00
Total			฿400.00
Take Payment			

POS App UI in Odoo ERP

302

Conference Chair (Aluminium)

4.00 x \$ 39.40 / Units

\$ 157.60

Large Desk

1.00 x \$ 1,799.00 / Units

\$ 1,799.00

Desk Organizer (Leather, L)

1.00 x \$ 5.10 / Units

\$ 5.10

Total\$ 1,961.70

Customer	Internal Note		Actions
1	2	3	Qty
4	5	6	%
7	8	9	Price
+/-	0	.	✕

Payment

Your logo

My Company (San Francisco)
Tel: +1 555-555-5556
info@yourcompany.com
http://www.example.com

Served by Mitchell Admin

302

Conference Chair (Aluminium)\$ 157.60

4.00 x \$ 39.40 / Units

Large Desk\$ 1,799.00

1.00 x \$ 1,799.00 / Units

Desk Organizer (Leather, L)\$ 5.10

1.00 x \$ 5.10 / Units

TOTAL\$ 1,961.70

Cash\$ 1,000.00

Card\$ 961.70

Generic App UI. List View.

 Point of Sale Dashboard Orders Products Reporting Configuration

Orders 

 Search...

☐ Order Ref	Session	Date	Point of Sale	Receipt Number	Custom
☐ Furniture Shop/0002	POS/00001	08/26/2025 20:44:31	Furniture Shop	Order 00003-001-0002	
☐ Furniture Shop/0001	POS/00001	08/26/2025 20:39:30	Furniture Shop	Order 00003-001-0001	
☐ ClosedDemo/0004	Furniture Shop/00001	08/24/2025 00:11:25	Furniture Shop	Order 00000-002-1002	
☐ ClosedDemo/0003	Furniture Shop/00001	08/24/2025 00:11:25	Furniture Shop	Order 00000-002-1001	
☐ ClosedDemo/0002	Furniture Shop/00000	08/24/2025 00:11:24	Furniture Shop	Order 00000-001-1002	
☐ ClosedDemo/0001	Furniture Shop/00000	08/24/2025 00:11:24	Furniture Shop	Order 00000-001-1001	

Generic App UI. Form View.

Point of Sale Dashboard Orders Products Reporting Configuration

Orders

Furniture Shop/0002 ⚙️

Pickings
1

Invoice Return Products

New Paid Posted

Order Ref Furniture Shop/0002

Date 08/26/2025 20:44:31

Session POS/00001

User ? Mitchell Admin

Customer

Fiscal Position

Products Payments Extra Info General Notes

Product	Lot/serial Nu...	Quantity	Unit Price	Disc.%	Taxes	Tax Excl.	Tax Incl.	🔍
[E-COM13] Conference Chair (Aluminium)		4.00	\$ 39.40	0.00		\$ 157.60	\$ 157.60	
[E-COM09] Large Desk		1.00	\$ 1,799.00	0.00		\$ 1,799.00	\$ 1,799.00	
[FURN_0001] Desk Organizer		1.00	\$ 5.10	0.00		\$ 5.10	\$ 5.10	

Taxes: \$ 0.00

Total: \$ 1,961.70

Total Paid (with rounding): \$ 1,961.70

Products	Payments	Extra Info	General Notes	
Date	Payment Method		Amount	
8/26/2025 20:44:24	Card			\$ 961.70
8/26/2025 20:44:18	Cash			\$ 1,000.00
Add a line				

UI Menu

The screenshot displays a software interface with a top navigation bar and a main content area. The navigation bar includes the following items: a barcode icon, 'Point of Sale', 'Dashboard', 'Orders' (highlighted with a blue border), 'Products', 'Reporting', and 'Configuration'. Below the navigation bar, the 'Orders' section is visible, featuring a gear icon. A dropdown menu is open under the 'Orders' tab, listing the following options: 'Orders', 'Sessions', 'Payments', 'Preparation Display', and 'Customers'. The main content area below the dropdown shows a table with two rows of data. The first row has a checkbox, 'Order Ref', and a 'Date' header. The second row has a checkbox, 'Furniture Shop/0002', and the date '08/26'.

☐	Order Ref	Date
☐	Furniture Shop/0002	08/26

Generic UI. Invoice Form.

New

Invoices

Draft Invoice INV/2025/00005 ⚙

Confirm

Cancel

Draft

Posted

Customer Invoice

INV/2025/00005

Customer Deco Addict
77 Santa Barbara Rd
Pleasant Hill CA 94523
United States – US12345673

Invoice Date 08/25/2025

Payment terms 30 Days

Invoice Lines

Other Info

Product	Quantity	Price	Taxes	Amount	
☰ [E-COM07] Large Cabinet	2.00	320.00		\$ 640.00	🗑
☰ [E-COM10] Pedal Bin	1.00	47.00		\$ 47.00	🗑
☰ Search a product	1.00	0.00		\$ 0.00	🗑

[Add a line](#)[Add a section](#)[Add a note](#)[Catalog](#)

Terms and Conditions

Untaxed Amount: **\$ 687.00**Total: **\$ 687.00**

Product List of Value (LoV) via List View

Search: Product



Product

Desk



Search...



1-12 / 12



Internal Refere...	Name	Variant Values	Sales Price	Cost	On Hand	Forecasted	
☆ E-COM06	Corner Desk Right Sit		147.00	600.00	0.00	0.00	
☆ E-COM09	Large Desk		1,799.00	1,299.00	-1.00	11.00	
☆ FURN_0001	Desk Organizer		5.10	0.00	-1.00	-1.00	

New

Close

UI Design Prototyping: Invoice Form View

Home

Invoice Form View

File Edit Users Master Transactions Reports Options

Close Save New Print

Invoice No.: INV25-001 Invoice Date: 07/01/2025

Customer Code: C100 Customer Name: Chaloem Sukh Co., Ltd.

Status: Draft

#	*Product Code	Product Name	*Quantity	*Unit Price	Units	Extended Price
1	DESK-11	Work Desk Wooden	1.00	6,000.00	PCS	6,000.00
2	CHAIR-2	Office Chair	2.00	2,000.00	PCS	4,000.00
3						

Total: 10,000.00

VAT: 700.00

Amount Due: 10,700.00

A **responsive UI (user interface)** automatically **adapts its layout and elements** to different **screen sizes, devices, and orientations** (desktop, tablet, mobile, landscape, portrait).

UI Design Prototyping: Invoice List Views



Search / Group /Filter: _____ [OK]

Next Action: ☒ Open Form (Primary Key) ☐ Select Record (Foreign Key)

Click Row to select a record: (Will open form or select the record, depending on “Next Action”)

Invoice No.	Date	Customer Name	Total Amount	VAT	Amount Due
INV24-721	27/12/2024	ABC Public Co., Ltd.	5,000.00	350.00	5,350.00
INV24-722	28/12/2024	AWN Public Co., Ltd.	1,000.00	70.00	1,070.00
INV24-723	28/12/2024	Sermmitr Advance Co., Ltd.	2,000.00	140.00	2,140.00
INV25-001	07/01/2025	Chaloem Sukh Co., Ltd	10,000.00	700.00	10,700.00

POS Sales Form View Generic Design (Minimalist)

POS Sales Form View Generic Design

User: Ying Sanrungroengkijthaveepop Role: Cashier

Sales Return Products Set Prices Logout Reports Authorization (App Controls)

[Close][New][Save][Clear][Print] (Form Controls)

Invoice No.: INV25-001 *Invoice Date: 07/01/2025 [Date]

Customer Code: Walk-in [LoV] Customer Name: No Name

#	*Product Code	Product Name		*Quantity		Unit Price	Units	Extended Price	Edit		
1	DESK-11	Work Desk Wooden		1.00		6,000.00	PCS	6,000.00	Edit	✕	▲
2	CHAIN-1	Link Chain 5mm		10.50		100.00	METER	1,050.00	Edit	✕	□
3	CHAIR-2	Office Chair		2.00		2,000.00	PCS	4,000.00	Edit	✕	
4	[LoV]		[-1]	1.00	[+1]						▼
Total:								11,050.00			

POS Sales List View Generic Design (Minimalist)

POS Sales List View

User: Ying Sanrungroengkijthaveepop Role: Cashier

Sales Return Products Set Prices Logout Reports Authorization (App Controls)

[Close][New]

Search / Group /Filter: _____ [OK]

Next Action: ☒ Open Form (Primary Key) ☐ Select Record (Foreign Key)

Click Row to select a record: (Will open form or select the record, depending on “Next Action”)

Invoice No.	Date	Customer Name	Total Amount
INV24-721	27/12/2024	No Name	5,000.00
INV24-722	28/12/2024	No Name	1,000.00
INV24-723	28/12/2024	No Name	2,000.00
INV25-001	07/01/2025	Chaloem Sukh Co., Ltd	10,000.00

Product List View Generic Design (Minimalist)

Product List View

User: Ying Sanrungroengkijthaveepop Role: Cashier

Sales Return Products Set Prices Logout Reports Authorization (App Controls)

[Close][New]

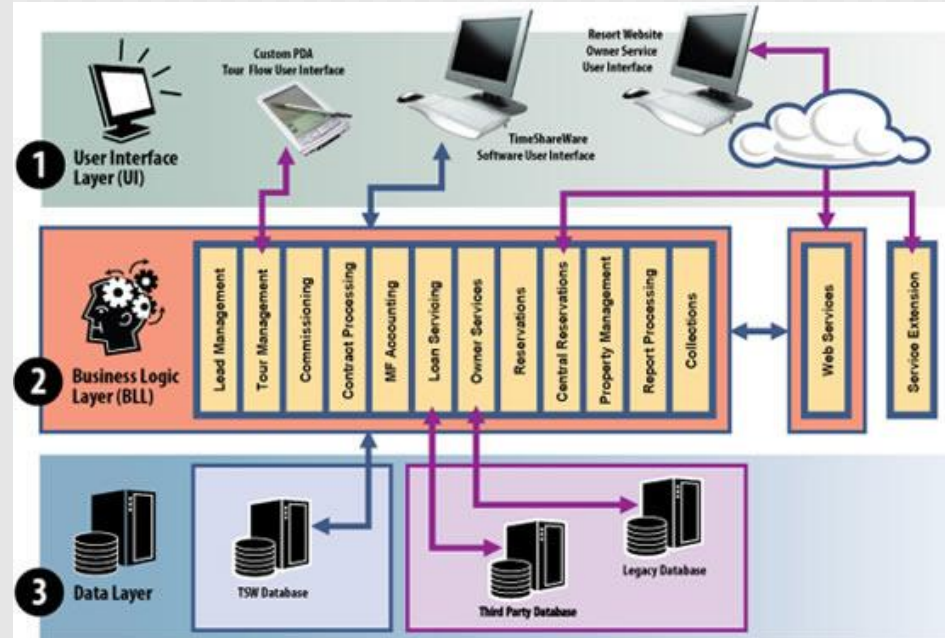
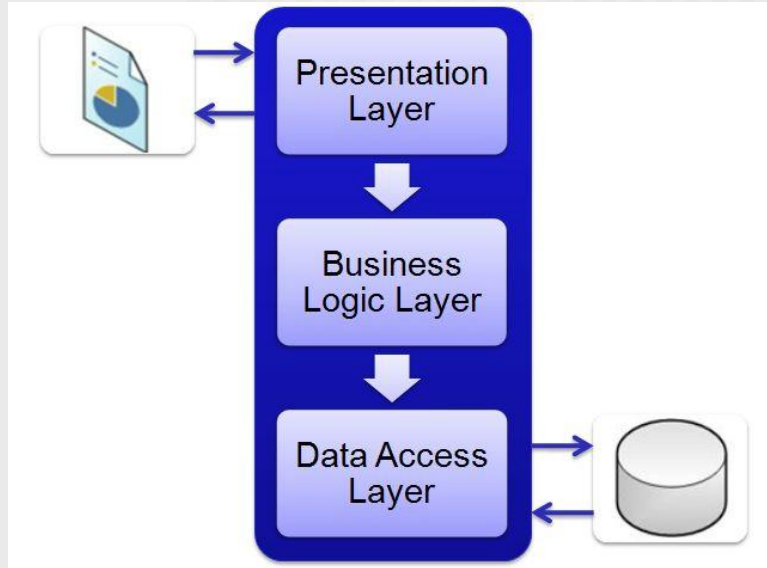
Search / Group /Filter: _____ [OK]

Mode: ☐ Action Menu (Primary Key) ☒ Select Record (Foreign Key)

Click Row to select a record: (Will open form or select the record, depending on “Next Action”)

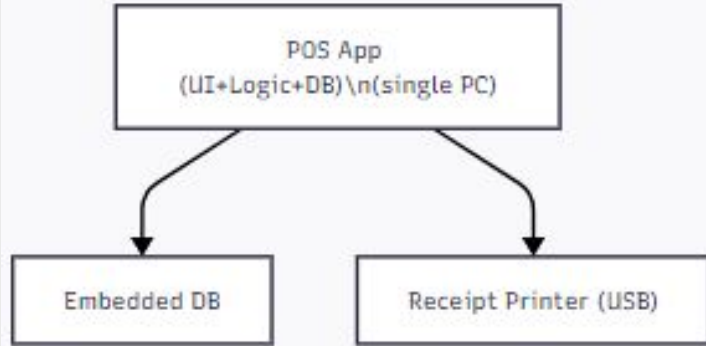
Product Code	Name	Unit Price	Units
CHAIN-1	Link Chain 5mm	100.00	METER
CHAIR-1	Dining Table Chair	1,500.00	PCS
CHAIR-2	Office Chair	2,000.00	PCS
COUCH-1	Leather sofa couch	20,000.00	PCS
DESK-11	Work Desk Wooden	6,000.00	PCS

The 3 Layers or Tiers: Data, Business Rules, and Interface

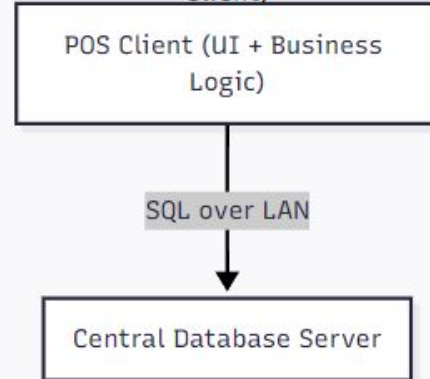


Compare POS Architecture Evolution

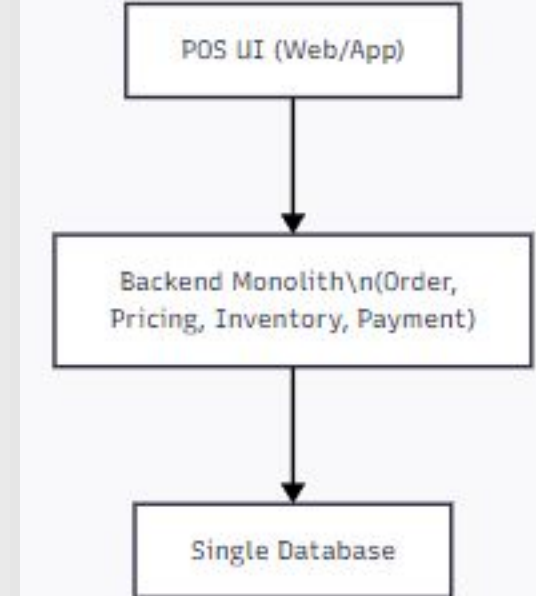
Singleton



Client-Server (2-tier, Fat Client)



Monolith (3-tier)



Presentation Layer vs. Service Layer

Presentation Layer

- **Purpose:** Handles **UI/UX** — what the user directly sees and interacts with.
- **Examples:**
 - Web front-end (HTML, React, Angular, etc.).
 - Mobile app GUI.
 - Desktop app forms/windows.

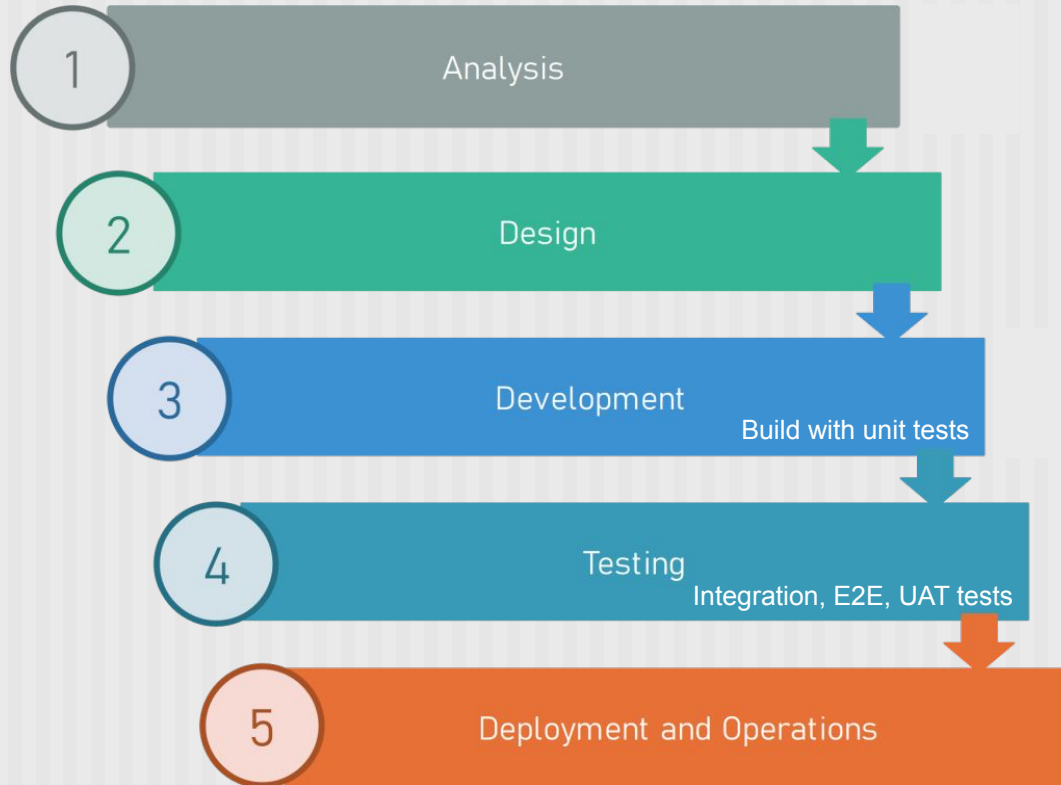
Key idea: Presentation = “face” of the system for “human” interaction.

Services Layer - “Headless”

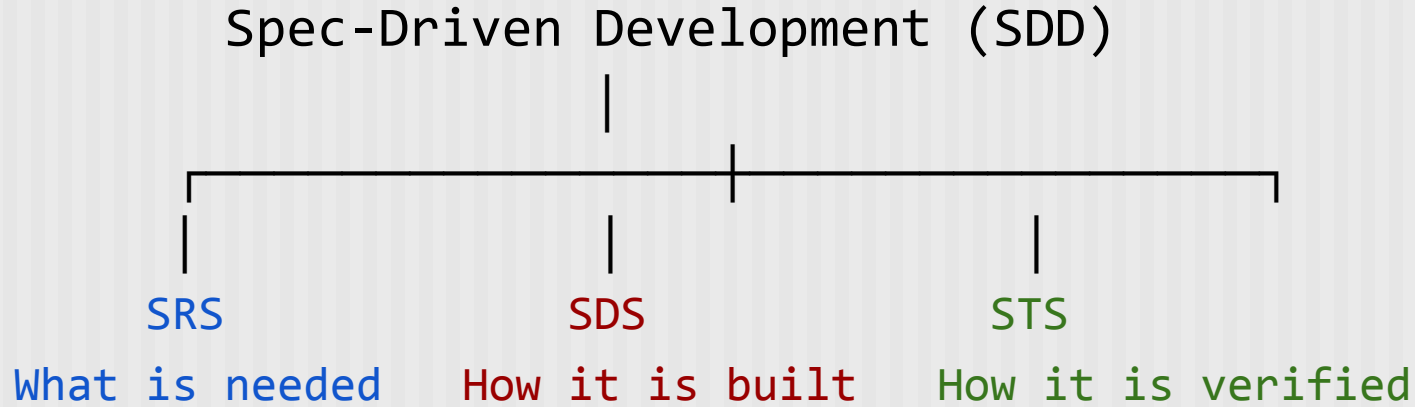
- **Purpose:** Provides **application-level business services** that can be consumed by **multiple presentation channels** (UI, mobile, API clients, even other systems).
- **Examples:**
 - REST API endpoints, SOAP services.
 - GraphQL resolver layer.
 - Service classes in a Java/Spring or .NET app that expose operations to controllers/UI.

Requirements Specification

Anatomy of a Software Project



Spec-Driven Development (SDD) for Coding Agents



SRS - software requirements specification

SDS - Software Design Specification

STS - Software Test Specification

Recommended Design Process

1. SRS (FR, BR, NFR)



2. Complete feature inventory (List: Feature-A, Feature-B, ...)



3. Requirements-to-features traceability (FR-17 is covered by Feature-A)



4. System-level SDS design (Architecture, tech stack, std. UI, API convention)



5. Feature-level SDS design specifications (Specs for each Feature-A, Feature-B, ...)



Implementation tasks and tests

Step 1: Requirements Analysis

Stakeholder Needs



User Requirements



Detailed System / Software Requirements

- ├ Functional Requirements
- ├ Non-Functional Requirements
- ├ Business Rules
- └ Constraints

Level	Example
Stakeholder requirement	“We need a system for employees to request IT support.” “The system must work on desktop, tablet, and mobile.”
User requirement	“A Requester must be able to create and view own Tickets.”
Functional requirement	“FR-01: The system shall allow a Requester to create a Ticket.”
Business rule	“BR-03: A new Ticket shall begin with status New.”
Non-functional requirement	“NFR-01: The UI shall be usable on desktop, tablet, and mobile.”
Constraint	“The frontend must use React + TypeScript.”

Should we separate BR from FR?

They can be bundled, but separation is usually cleaner.

Functional Requirement (FR) - What the system does.

Business Rule (BR) - A rule, constraint, policy, or condition governing that behavior.

Example:

- **FR-08:** The system shall allow a Requester to upload attachments to a Ticket.
- **BR-12:** Maximum attachment size is 5 MB.
- **BR-13:** Maximum five active attachments per Ticket.
- **BR-14:** Removed attachments must not be downloadable.

One large FR may contain many testable rules, making it harder to trace, review, change, and test.

Guideline

- Put the main capability in the FR.
- Put reusable or independently testable policy rules in BRs. Avoid creating a BR for every tiny detail.

The Software Requirements Specification (SRS) Document

$$\text{SRS} = \text{BR} + \text{FR} + \text{NFR} + \text{supporting context}$$

The requirement catalogue

BR Business goals and outcomes

FR Functions and system behavior

NFR Quality attributes and constraints

These requirements form the core of the SRS.

Purpose & scope

Business context, objectives, system boundary, and exclusions.

Actors & interfaces

Users, external systems, inputs, outputs, and interactions.

Assumptions & constraints

Dependencies, operating conditions, policies, and technical limits.

Acceptance & traceability

Testable criteria and links from requirements to design and tests.

Requirements List (minimalist)

1. **Serve customers at the counter**
 - Cashiers can start a checkout, add or remove items, apply valid discounts, and see the running total.
 - A sale can be **cancelled** any time before it is finished.
2. **Accept payments**
 - Customers can pay **by cash** or **by credit card** which is processed via a **Payment Service Provider** (eg., K-Bank)
 - The system **works with the card machine** from, say, K-Bank to send the amount and receive approval
 - If a card is **declined**, the cashier can retry, switch to cash, or cancel the sale.
3. **Print documents: Receipt** for successful sale, **Refund Receipt** for refunds. Prints to store **receipt printer**.
4. Updates stock in **store's inventory service** app: **reduces** stock of the sold item, adds back stock of returned items.
5. **Handle returns and refunds.**
 - Cashiers or managers can find the original sale and **process a return**.
 - The system **refunds** the customer (cash or card, following store policy) and records the reason.
6. **Manager Functions**
 - Managers can view a **daily sales summary** for a selected date, including total cash & card amount received.
 - Managers can **add or update products and prices** to be used by POS.

SRS Prompt Session Example



<https://docs.google.com/document/d/1Y87HZ0RwSAUjcm8pPtiZ-s5-iOuZmo7L3uZXpuleovI/edit?usp=sharing>

For this chat, assume I'm a software engineering student wanting your help in creating good SE artifacts such as Requirements Specifications, Design Specs, Test Scenarios, etc.

I have these requirements provided to me for a POS system by my instructor. I have to iteratively refine it until they become numbered FR, BR, and NFR. First, please read this and ask me questions about parts of this POS system that may need more clarification before you can draft the requirements lists.

1. **Serve customers at the counter**
 - Cashiers can start a checkout, add or remove items, apply valid discounts, and see the running total.
 - A sale can be **cancelled** any time before it is finished.
2. **Accept payments**
 - Customers can pay **by cash** or **by credit card** which is processed via a **Payment Service Provider** (eg., K-Bank)
 - The system **works with the card machine** from, say, K-Bank to send the amount and receive approval
 - If a card is **declined**, the cashier can retry, switch to cash, or cancel the sale.
3. **Print documents: Receipt** for successful sale**, Refund Receipt** for refunds. Prints to store **receipt printer**.
4. Updates stock in **store's inventory service** app**: reduces** stock of the sold item, adds back stock of returned items.
5. **Handle returns and refunds. **
 - Cashiers or managers can find the original sale and **process a return**.
 - The system **refunds** the customer (cash or card, following store policy) and records the reason.
6. **Manager Functions**
 1. Managers can view a **daily sales summary** for a selected date, including total cash & card amount received.
 2. Managers can **add or update products and prices** to be used by POS.

Software Design

1. Starts with the Software Architecture
2. Decompose into parts/component Features
3. What are the functions of, and interfaces between components?
4. Resulting document: Software Design Specification (SDS)

Contents of SDS Document: System-Level

I. **System-Level SDS.** This defines shared technical decisions that apply across the application:

- System architecture
- Technology stack
- Major components and modules
- Overall data model
- Authentication and authorization approach
- UI Design System Specification
- API conventions
- Error-handling conventions
- Security, performance, logging, and other cross-cutting NFRs

Contents of SDS Document: Feature Level

II. For **Each Feature** , create a specification containing:

1. Feature purpose and scope
2. Related SRS requirements
3. Workflow and state transitions
4. Business rules
5. Data model changes
6. API endpoints
7. UI behavior
8. Permissions
9. Validation and error handling
10. Acceptance criteria
11. Test considerations

UI Design Specifications

- Theme and color palette
- Semantic color usage, such as success, warning, error, and information
- Typography and heading hierarchy
- Spacing and layout grid
- Buttons, forms, tables, cards, and other shared components
- Icons and imagery
- Responsive breakpoints
- Accessibility requirements, including contrast and keyboard navigation
- Light/dark mode, if supported
- Standard loading, empty, confirmation, and error states

Document	Example
SRS/NFR	NFR-UI-01: The application shall conform to the company's corporate branding guidelines.
SRS/NFR	NFR-UI-02: All text and interactive controls shall meet WCAG 2.1 AA contrast requirements.
SDS	Primary color: #0057B8; secondary color: #F5A623; error color: #D32F2F.
SDS	Primary buttons use the corporate blue background with white text.
SDS	Headings use the corporate typeface; body text uses Arial as the fallback.

KMUTT Color Theme

https://www.kmutt.ac.th/en/corporate_identity/colors/

COLORS



PANTONE
172 C

C 0 M 85 Y 100 K 0
R 255 G 70 B 18



PANTONE
123 C

C 0 M 25 Y 90 K 0
R 255 G 198 B 39



PANTONE
415 C

C 46 M 38 Y 38 K 2
R 145 G 145 B 145

The Colors of the University

The Colors of the University are orange-yellow which signify dynamic movement and strength. Both colors also activate enthusiasm. These colors are comparable to dynamic movement of the apparatus and machinery which is a crucial part of the teaching and learning in the university.

Color schemes for website and application. :

Yellow : #FFC72C

Orange : #FA4616

Blue Grey : #7B8189

สีเหลือง

Yellow #FFC72C

สีแสด

Orange #FA4616

สีเทา

Blue Grey #7B8189

Recommended Design Process

1. SRS (FR, BR, NFR)



2. Complete feature inventory (List: Feature-A, Feature-B, ...)



3. Requirements-to-features traceability (FR-17 is covered by Feature-A)



4. System-level SDS design (Architecture, tech stack, std. UI, API convention)



5. Feature-level SDS design specifications (Specs for each Feature-A, Feature-B, ...)



Implementation tasks and tests

2. Complete Feature Inventory: POS

ID	Feature	Feature scope
Feature-A	User Authentication and Access Control	Sign-in, user roles, function permissions, and store-based access restrictions for Cashiers, Managers, Administrators, and Head-Office Managers.
Feature-B	Store and Terminal Operating Context	Identification of the current store and POS terminal, association of transactions with that context, and support for multiple stores and terminals.
Feature-C	Product Catalog and Price Management	Addition and modification of products, product information, VAT-inclusive selling prices, and manager access restrictions.
Feature-D	Sale Lifecycle Management	Starting a sale, maintaining its current state, cancelling it before completion, and preventing modification after completion.
Feature-E	Cart and Sale-Item Management	Adding products, changing quantities, removing products, and recalculating the sale when its items change.
Feature-F	VAT-Inclusive Pricing and Total Calculation	Calculation and display of item amounts, subtotal, Thai VAT information, discount, and final sale amount using Thai baht and satang.

2. Complete Feature Inventory: POS

ID	Feature		Feature scope
Feature-G	Order-Level Management	Discount	Discount Percentage and Discount Amount entry, percentage-to-amount calculation, satang rounding, manual discount-amount handling, validation, and recalculation when the cart changes.
Feature-H	Payment Selection	Method	Selection of either cash or credit card, enforcement of the no-split-payment rule, and permitted switching of payment methods.
Feature-I	Cash Payment Processing		Entry of cash tendered, validation that sufficient cash was received, change calculation, and cash-sale completion.
Feature-J	Card Authorization	Payment	Creation of a unique payment-attempt reference, durable recording of the attempt, communication with the card machine and Payment Service Provider, and handling of approved or declined results.
Feature-K	Card Payment Recovery and Reconciliation		Handling communication interruptions, unresolved payment results, approved payments that could not be finalized, PSP status queries, duplicate-charge prevention, and recovery of affected sales.
Feature-L	Sale Finalization and Durable Recording		Validation of successful payment, assignment of a unique sale identifier, permanent recording of the sale, and atomic creation of the associated pending inventory update.

2. Complete Feature Inventory: POS

ID	Feature	Feature scope
Feature-M	Receipt Printing and Reprinting	Printing a receipt after successful sale completion, reporting printer failures, and reprinting without duplicating or reversing the sale.
Feature-N	Inventory Update Outbox	Creation and durable storage of a pending inventory-update record containing the sale, store, terminal, date and time, product, and quantity information.
Feature-O	Inventory Message Delivery	Asynchronous message transmission, acknowledgement processing, retained-message retry, and duplicate inventory-update prevention through unique transaction identifiers.
Feature-P	Operational Exception Monitoring	Viewing unresolved card payments, payment-recovery cases, pending inventory messages, and inventory messages that have repeatedly failed delivery.
Feature-Q	Daily Sales Reporting	Selection of a business date, assigned-store reporting for Managers, selected-store reporting for Administrators and Head-Office Managers, consolidated reporting, and separate cash and card totals.
Feature-R	Audit Trail	Recording the responsible user, store, terminal, date, and time for completed sales, cancellations, discounts, card-payment recovery, and product or price changes.

Recommended Design Process

1. SRS (FR, BR, NFR)



2. Complete feature inventory (List: Feature-A, Feature-B, ...)



3. Requirements-to-features traceability (FR-17 is covered by Feature-A)



4. System-level SDS design (Architecture, tech stack, std. UI, API convention)



5. Feature-level SDS design specifications (Specs for each Feature-A, Feature-B, ...)



Implementation tasks and tests

3. Requirements-to-features traceability



Lecture 3 ChatGPT Sessions

<https://docs.google.com/document/d/1Y87HZ0RwSAUjcm8pPtiZ-s5-iOuZmo7L3uZXpuleovI/edit?usp=sharing>

4. System-level SDS design



Kaching System Level SDS v0.1.docx

https://docs.google.com/document/d/1FHjOQHuzUGmqe20xRl5IDQNjBWumTP7T/edit?usp=drive_link&oid=108394895055428723586&rtpof=true&sd=true

Before calling it **v1.0 Baseline** , we should close these items:

- Add the proposed requirements FR-039–FR-043, BR-021, and NFR-012–NFR-016 to the SRS.
- Update the requirements-to-features traceability matrix accordingly.
- Replace [Student Name] and add approval/reviewer information.
- Confirm production-specific details that are currently represented by architectural assumptions:
 - Exact K-Bank ECR and PSP protocols, Receipt-printer model and communication protocol, RabbitMQ hosting provider
 - Google Cloud region, Inventory message schema and acknowledgement rules
- Obtain stakeholder approval for the architectural decisions and risks.

The following are intentionally not included because they belong in **later feature-level design** specifications:

- Complete REST endpoint specifications
- Detailed database schema and Prisma models
- Screen designs and interaction flows
- Exact receipt layout
- Class/component designs
- Detailed test scenarios
- Deployment scripts and infrastructure code



TikTockT-System-Level-SDS-Draft.docx



TokTickT-System-Level-SDS-v1.0.docx

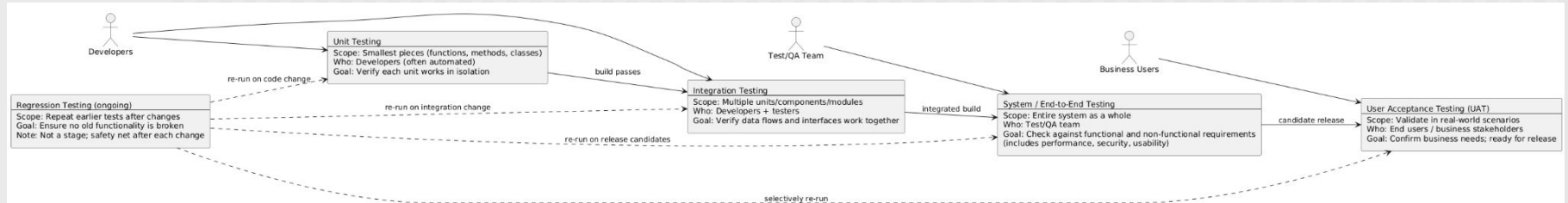
5. Feature-level SDS design specifications

Code Generation (Build & Unit Test)

1. Focus on individual components
2. Goal: a working, flexible, robust, ... piece of software

Testing Types in Large Systems

Type of Testing	Scope	Who Runs It	Goal
1. Unit Testing	Smallest pieces of code (functions, methods, classes)	Developers (often automated)	Verify each unit works in isolation
2. Integration Testing	Multiple units/components/modules interacting	Developers + Testers	Verify data flows and interfaces work correctly together
3. System Testing / End-to-End (E2E)	The entire system as a whole	Testers	Check functional and non-functional requirements (performance, security, etc.)
4. User Acceptance Testing (UAT)	Business users validate the system in real-world scenarios	End Users + Testers	Confirm the system meets business needs and is ready for release. Get users to sign off.
Regression Testing	Repeat earlier tests (unit, integration, system) after changes	Testers	It's not a stage. Another, optional check after each test "stage" above.



Testing Types in Large Systems



Unit Testing

Scope: Smallest pieces (functions, methods, classes)

Who: Developers (often automated)

Goal: Verify each unit works in isolation

re-run on code change →

Regression Testing (ongoing)

Scope: Repeat earlier tests after changes

Goal: Ensure no old functionality is broken

Note: Not a stage; safety net after each change

re-run on integration change

Other Tests

Other Common Specialized Test Types (fitting around System Testing)

- **Performance / Load Testing** – can the system handle expected volume?
- **Security / Penetration Testing** – are vulnerabilities closed?
- **Compatibility Testing** – does it work across devices, browsers, OS?
- **Recovery / Failover Testing** – does it handle crashes gracefully?

Maintenance and Software Operations

1. Correcting errors found after the software has been delivered
2. Making changes not anticipated via the regular UI
3. Adapting the software to changing requirements, changing environments, ...

Lab 2. Requester Ticketing MVP with UI Foundations

CPE 334 Introduction to Software Engineering in the Age of AI
Sections 1, 2, HS, 31 and 32. Semester: 1/2026.

Lab 2. TokTickIT Requester Ticketing MVP with UI Foundations

Figure 1. Illustrative TokTickIT Ticket Detail screen using the Zen Green template design language. Lab 2 implements the Requester-facing Create Ticket, My Tickets, Ticket Detail, and Attachment functions. Later labs will add role-specific IT Staff controls, communication, workflow, and Actions Taken.

TokTickIT My Tickets Create Ticket Profile

My Tickets > Ticket Details Back to My Tickets

Ticket No.	Ticket Date	Category	Related System
TKT-2025-001234	May 12, 2025 09:14 AM	Hardware	Corporate Laptop
Requester	Requested Priority	IT Priority	Current Status
Jennifer Anderson	Medium	Medium	In Progress
Ticket Owner	Summary		
Michael Brown (IT Support)	Laptop battery drains quickly		
Description			
My laptop battery is draining much faster than usual even when the system is idle. This started happening after last week's Windows update.			
Resolution Summary			
No resolution summary available yet.			

Public Comments 3 Attachments 2 Service Actions 1 Event Log 6

Add Comment

Type your comment here... Post Comment

JA Jennifer Anderson Requester May 13, 2025 11:45 AM
Thank you for the update. Please let me know if you need any additional information.

MB Michael Brown IT Support May 13, 2025 10:30 AM
We are investigating the issue on your device. We'll update you shortly.

JA Jennifer Anderson Requester May 12, 2025 09:20 AM
Just adding that this issue occurs even when I close all applications.

THE END

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